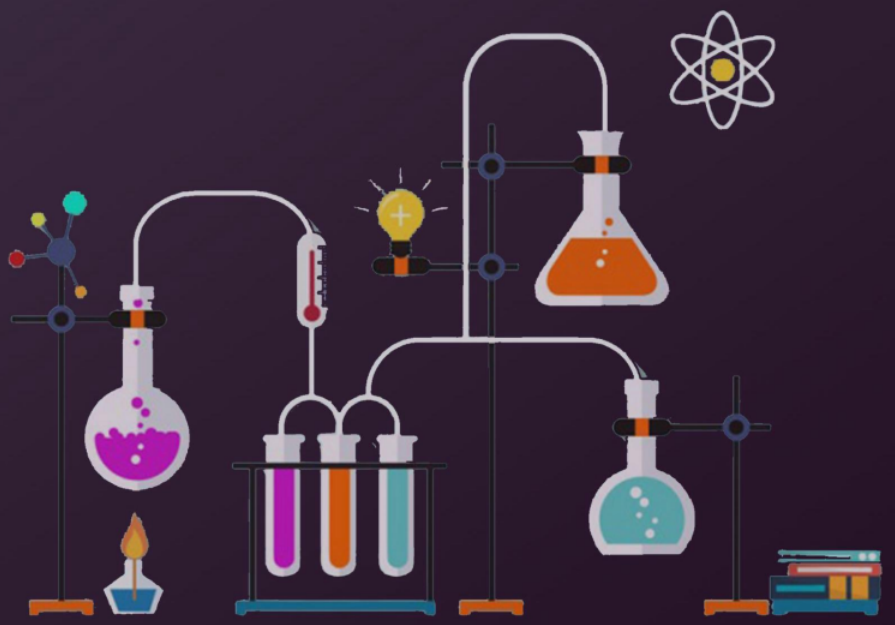
CHEMISTRY

Ionic Equilibrium



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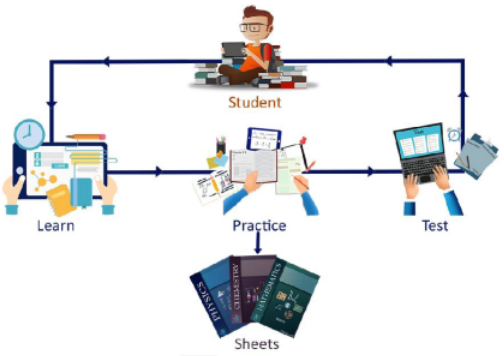
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Electrolytes and Degree of Dissociation	1 - 2	24,25	1-4	
2	Acid, Basis and Salts	3 - 18	1,2, 22	5,6	7
3	Lewis Concept and Normality		26	7	5,13-15
4	pH Scale and Properties of Water	19 - 26	3,4, 6-8,27	8-16	
5	Calculation of pH - Part 1	27 - 30		17-19	9
6	Calculation of pH - Part 2	31 - 39		20-22	
7	Calculation of pH - Part 3	40 - 49		23,24	
8	Summary of pH Calculation			25,26	
9	Hydrolysis of Salts		18	27-29	6
10	Hydrolysis of [WA - SB & WA-WB] Type Salt		19	30-33	10
11	Hydrolysis - The Energy Provider	50			
12	Buffer Solutions	51 - 55	9-14,21	34-38	8
13	Summary of pH, hydrolysis and Buffer	56	20,28	39,40	
14	Indicators	57 - 64		41,42	2-4
15	Solubility		5, 15-17, 23,29,30		
16	Solubility and Solubility Product	65 - 73	31	43-50	11,12
17	Important JEE Advanced 2018 Problem	74 - 75			
18	ACC	76 - 83		51-55	

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Electrolytes and Degree of Dissociation

Q1. Basic strength of NH_4OH in presence of NH_4Cl
(A) Increases
(B) Remains unchanged
(C) Decreases
(D) Some times increases or sometimes decreases

Q2. No. of total ion's and resultant total charge in A_2B_3 electrolyte's :-
(A) Five and + 1 (B) Five and - 1
(C) 0 and 0 (D) Five and Zero

Acid, Bases and Salts

Q3. Which of the following is an acidic salt :-
(A) Na_2S (B) Na_2SO_3
(C) $NaHSO_3$ (D) Na_2SO_4

Q4. Select the species which can function as - Lewis base, bronsted acid and bronsted base:-
(a) H_2O (b) NH_4^+ (c) N^{-3}
Correct code is :-
(A) Only a (B) a, b
(C) a, c (D) b, c

Q5. The conjugated base of $(CH_3)_2\overset{+}{N}H_2$ is :-
(A) CH_3NH_2 (B) $(CH_3)_2N^+$
(C) $(CH_3)_2N$ (D) $(CH_3)_2NH$

Q6. Water is a :-
(A) Protogenic solvent
(B) Protophilic solvent
(C) Amphiprotic solvent
(D) Aprotic solvent

Q7. Basicity of H_3PO_3 and H_3PO_2 are respectivily-
(A) 1 and 2 (B) 2 and 3
(C) 3 and 2 (D) 2 and 1

Q8. NH_3 gas dissolves in water to give NH_4OH, in this reaction, water acts as :-
(A) An acid
(B) A base
(C) A salt
(D) A conjugate base

Q9. Conjugate acid of $Zn(OH)_2$ is :-
(A) $Zn(OH)^+$ (B) $Zn(OH_3)^-$
(C) Zn^{2+} (D) None

Q10. In the reaction
$HNO_3 + H_2O \rightleftharpoons H_3O^+ + NO_3^-$,
the conjugate base of HNO_3 is :-
(A) H_2O
(B) H_3O^+
(C) NO_3^-
(D) H_3O^+ and NO_3^-

Q11. Addition of HCl will not suppress the ionization of-
(A) acetic acid
(B) Benzoic acid
(C) H_2S
(D) Sulphuric acid

Q12. The basicity of phosphorous acid is-
(A) 1 (B) 2
(C) 3 (D) 4

Q13. Gaseous hydrogen chloride is a very poor conductor of electricity but a solution of hydrogen chloride in water is a good conductor. This is due to the fact that :-

(A) Water is a good conductor of electricity

(B) Hydrogen chloride ionises in water

(C) A gas cannot conduct electricity but a liquid can

(D) HCl does not obey Ohm's law where as the solution does

Q14. In the process : $NH_3 + NH_3 \rightleftharpoons NH_2^- + NH_4^+$, The nature of ammonia is:-

(A) Acidic (B) Basic

(C) Amphoteric (D) None

Q15. Lewis base is :-

(A) NH_3 (B) CO_2

(C) BCl_3 (D) K^+

Q16. Sulphanilic acid is a/an

(A) Arrhenius acid

(B) Lewis base

(C) Neither (A) or (B)

(D) Both (A) and (B)

Q17. Which of the following is not an acidic salt:-

(A) $NaHSO_4$ (B) HCOONa

(C) NaH_2PO_3 (D) None of them

Q18. The conjugate base of $H_2PO_4^-$ is :-

(A) HPO_4^{2-} (B) P_2O_5

(C) H_3PO_4 (D) PO_4^{3-}

pH Scale and Properties of Water

Q19. Least pKa value is for this acid :-

(A) F_3CCOOH

(B) CH_3COOH

(C) $(CH_3)_2CHCOOH$

(D) $ClCH_2.COOH$

Q20. Which of the following has lowest pH value:-

(A) 1 M HCl (B) 1 M NaOH

(C) 1 M H_2SO_4 (D) 1 M C_2H_5OH

Q21. If pH = 3.31, then find out $[H^+]$ (Approximately)

(A) 3.39×10^{-4} (B) 5×10^{-4}

(C) 3.0×10^{-3} (D) None

Q22. At 25°C K_b for BOH = 1.0×10^{-12}. 0.01 M solution of BOH has $[OH^-]$:-

(A) 1.0×10^{-6} M (B) 1.0×10^{-7} M

(C) 1.0×10^{-5} M (D) 2.0×10^{-6} M

Q23. What is the OH^- concentration of an aqueous solution of a substance whose pH = 3·2 :-

(A) $10^{-3\cdot8}$ (B) $10^{-3\cdot2}$

(C) $10^{-10\cdot8}$ (D) $10^{-10\cdot2}$

Q24. Ionic product of water will increase, if :-

(A) Dissociation the pressure

(B) Add H^+

(C) Add OH^-

(D) Increase the temperature

Q25. Ionic product of water is equal to :-

(A) Dissociation constant of water × $[H_2O]$

(B) Dissociation constant of water × $[H^+]$

(C) Product of $[H_2O]$ and $[H^+]$

(D) Product of $[OH^-]^2$ and $[H^+]$

Q26. An aqueous solution whose pH = 0 is :-

(A) Basic (B) Acidic

(C) Neutral (D) Amphoteric

Calculation of pH-Part-1 (Strong Acid / Base)

Q27. If pH value of A, B, C and D are 9.5, 2.5, 3.5 and 5.5 respectively, then the strongest acid is

(A) A (B) B (C) C (D) D

Q28. Which of the following resultant solution has pH=1 :-

(A) 100 ml $\frac{N}{100}$ HCl + 100 ml $\frac{M}{100}$ NaOH

(B) 55 ml $\frac{M}{10}$ HCl + 45 ml $\frac{M}{10}$ NaOH

(C) 10 ml $\frac{M}{10}$ HCl + 90 ml $\frac{M}{10}$ NaOH

(D) 75 ml $\frac{M}{5}$ HCl + 25 ml $\frac{M}{5}$ NaOH

Q29. The pH of a solution is 7.00. To this solution sufficient base is added to increase the pH to 12.0. The increase OH^- ion concentration is-

(A) 5 times (B) 1000 times

(C) 10^5 times (D) 4 times

Q30. Stomach acid is a solution of HCl with concentration of 1.2×10^{-3} M. What is the pH of stomach acid-

(A) 3.92 (B) 2.92

(C) 4.92 (D) 1.92

Calculation of pH Part-2 (Weak Acid / Base)

Q31. $HCOOH \rightleftharpoons HCOO^- + H^+$; $K_a = 1.7 \times 10^{-4}$

The ionization of formic acid is represented above. Calculate $[H^+]$ of a solution initially containing 0.1 M HCOOH and 0.05 M HCOONa.

(A) 8.5×10^{-5} M (B) 3.4×10^{-4} M

(C) 4.1×10^{-3} M (D) 1.8×10^{-2} M

Q32. Which of the following salts undergoes hydrolysis in water:-

(A) Na_3PO_4

(B) CH_3COONa

(C) $NaNO_3$

(D) Both of (A) and (B)

Q33. The degree of dissociation of 0.1 M HCN solution is 0.01%. Its ionisation constant would be :-

(A) 10^{-3} (B) 10^{-5}

(C) 10^{-7} (D) 10^{-9}

Q34. The pH of a 0.1 M aqueous solution of a very weak acid (HA) is 3. What is its degree of dissociation?

(A) 1% (B) 10%

(C) 25% (D) 0.1%

Q35. Find out the value of 'α' of 10^{-2} M HCN for which is given at equilibrium concentration of $[H^+] = 10^{-3}$

(A) 5% (B) 10%

(C) 15% (D) 20%

Q36. What is the ionization constant of a weak monobasic acid if the hydronium ion concentration of a 0.40 M acid solution is 1.40×10^{-4} M ?

(A) 1.96×10^{-8} (B) 1.22×10^{-9}

(C) 4.90×10^{-8} (D) 1.40×10^{-6}

Q37. The pH of a 0.01 M solution of acetic acid is

(A) 2

(B) Less than 2

(C) Greater than 2

(D) Cannot be predicted

Q38. The K_a values of formic acid and acetic acid are respectively 1.75×10^{-4} and 1.75×10^{-5}. The ratio of acid strength of 0.1 N acids is:-

(A) 10 (B) 3.16

(C) 4.16 (D) None

Q39. Which of the following solutions has the maximum pH value ?

(A) 0.2 M HNO_3

(B) 0.2 M HCl

(C) 0.2 M CH_3COOH

(D) 0.2 M CH_3COONa

Calculation of pH- Part-3 (Hydrolysis of Salts)

Q40. A strong acid solution whose pH = 4 is mixed with strong base solution whose pH = 10 then the solution after mixing is :-

(A) strongly basic (B) strongly acidic

(C) neutral (D) Weakly acidic

Q41. The correct order for the increasing extent of hydrolysis is-

(A) $PO_4^{3-} < HPO_4^{2-} < H_2PO_4^-$

(B) $H_2PO_4^- < HPO_4^{2-} < PO_4^{3-}$

(C) $HPO_4^{2-} < PO_4^{3-} < H_2PO_4^-$

(D) $PO_4^{3-} < H_3PO_4^- < HPO_4^{2-}$

Q42. If the K_b value in the hydrolysis reaction, $B^+ + H_2O \square BOH + H^+$; is 1.0×10^{-6}, then the hydrolysis constant of the salt would be:

(A) 1.0×10^{-6} (B) 1.0×10^{-7}

(C) 1×10^{-8} (D) 1.0×10^{-9}

Q43. Which is the correct alternate for hydrolysis constant of NH_4CN ?

(A) $\sqrt{\frac{K_w}{K_a}}$ (B) $\frac{K_w}{K_a \times K_b}$

(C) $\sqrt{\frac{K_b}{K_c}}$ (D) $\frac{K_a}{K_b}$

Q44. Which of the following salts undergoes hydrolysis :-

(A) CH_3COONa (B) KNO_3

(C) NaCl (D) K_2SO_4

Q45. pH of water is 7. When any substance Y is dissolved in water then pH becomes 13. Substance Y is a salt of :-

(A) Strong acid and strong base

(B) Weak acid and weak base

(C) Strong acid and weak base

(D) Weak acid and strong base

Q46. What is the percentage degree of hydrolysis (approx.) of KCN in $\frac{N}{80}$ solution when the dissociation constant for HCN is 1.6×10^{-9} and $K_w = 10^{-14}$.

(A) 5.26 (B) 2.24

(C) 8.2 (D) 4.1

Q47. Which of the following salts undergoing anionic hydrolysis :-

(A) $CuSO_4$ (B) NH_4Cl

(C) $AlCl_3$ (D) K_2CO_3

Q48. The degree of dissociation of pure water at 25°C is found to be 1.8×10^{-9}. Find K_w and K_d at 25°C.

(A) 3.24×10^{-18}; 5.83×10^{-2}

(B) 1×10^{-14}; 1.8×10^{-16}

(C) 1.8×10^{-16}; 1×10^{-14}

(D) 1×10^{-14}; 1×10^{-14}

Q49. The degree of hydrolysis of a salt of weak acid and a weak base

(A) Increases with concentration

(B) Independent with temperature

(C) Is independent of concentration

(D) decreases with concentration

Hydrolysis - The Energy Provider

Q50. Which is incorrect about the hydrolysis of CH_3COONa ?

(A) Anionic hydrolysis

(B) $pH = \frac{1}{2}(pKw + pKa + \log c)$

(C) $[H^+] = \sqrt{\frac{C \times K_w}{Ka}}$

(D) $K_h = \frac{K_w}{Ka}$

Buffer Solutions

Q51. To a 50 mL of 0.05M formic acid how much volume of 0.10M sodium formate must be added to get a buffer solution of pH = 4.0 ? (pK_a of the acid is 3.8)

(A) 50 mL (B) 4 mL

(C) 39.6 mL (D) 100 mL

Q52. The buffer solution play an important role in :-

(A) Increasing the pH value

(B) Decreasing the pH value

(C) Keeping the pH constant

(D) Solution will be neutral

Q53. The pK_a of formic acid is 3.75. The pH of a solution containing 0.02 M formic acid and 0.01 M sodium formate is-

(A) 3.80 (B) 5.00

(C) 3.45 (D) 4.25

Q54. What will be the pH of an aqueous solution of 1.0 M ammonium formate ?

Given : $pK_a = 3.8$ and $pK_b = 4.8$

(A) 7.5 (B) 3.4

(C) 6.5 (D) 10.2

Q55. In a buffer solution the ratio of concentration of NH_4Cl and NH_4OH is 1 : 1 when it changes in 2 : 1 what will be the value of pH of buffer :-

(A) Increase (B) Decrease

(C) No effect (D) N.O.T.

Summary of pH , hydrolysis and Buffer

Q56. A solution contains 0.2M NH_4OH and 0.2M NH_4Cl. If 1.0 mL of 0.001 M HCl is added to it. What will be the $[OH^-]$ of the resulting solution $[K_b = 2 \times 10^{-5}]$:-

(A) 2×10^{-5} (B) 5×10^{-10}

(C) 2×10^{-3} (D) None of these

Indicators

Q57. What is the suitable indicator for titration of NaOH and oxalic acid:-

(A) Methyl orange (B) Methyl red

(C) Phenolphthalein (D) Starch solution

Q58. Which indicator works in the pH range 8 – 9.8 .

(A) Phenolphthalein (B) Methyl orange

(C) Methyl red (D) Litmus

Q59. Phenolphthalein is not a good indicator for titrating

(A) NaOH against oxalic acid

(B) NaOH against HCl

(C) NaOH against H_2SO_4

(D) Ferrous sulphate against $KMnO_4$

Q60. In bases, methyl orange turns :-

(A) Red (B) Pink

(C) Yellow (D) Colourless

Q61. 5 ml of N HCl, 20 ml of N/2 H_2SO_4 and 30 ml of N/3 HNO_3 are mixed together and volume made to one litre. The normally of the resulting solution is :-

(A) $\frac{N}{5}$ (B) $\frac{N}{10}$

(C) $\frac{N}{20}$ (D) $\frac{N}{40}$

Q62. The pH indicators are-

(A) salts of strong acids & strong bases

(B) salts of weak acids & weak bases

(C) either weak acids or weak bases

(D) either strong acids or strong bases

Q63. What fraction of an indicator Hln is in the basic form at a pH of 6 if pK_a of the indicator is 5 ?

(A) $\frac{1}{2}$ (B) $\frac{1}{11}$ (C) $\frac{10}{11}$ (D) $\frac{1}{10}$

Q64. The degree of hydrolysis of 0.1 M solution of conjugate base of HA is 0.01. Find the H^+ concentration in 0.4 M solution of A^-.

(A) 5×10^{-11} M (B) 5×10^{-12} M
(C) 2×10^{-3} M (D) 2×10^{-4} M

Solubility and Solubility Product

Q65. Let the solubilities of AgCl in H_2O, 0.01 M $CaCl_2$, 0.01 M NaCl & 0.05 M $AgNO_3$ be S_1, S_2, S_3 & S_4 respectively what is the correct relationship between these quantities. Neglect any complexation.

(A) $S_1 > S_2 > S_3 > S_4$ (B) $S_1 > S_2 = S_3 > S_4$
(C) $S_1 > S_3 > S_2 > S_4$ (D) $S_4 > S_2 > S_3 > S_1$

Q66. How many grams of CaC_2O_4 will dissolved in distilled water to make one litre of saturated solution ($K_{sp} = 2.5 \times 10^{-9}$ and its molecular mass is 128.)

(A) 0.0064 g (B) 0.0128 g
(C) 0.0032 g (D) 0.0640 g

Q67. Solubility products of $M(OH)_3$ and $M(OH)_2$ are 10^{-23} and 10^{-14} respectively. Which will be precipitated first on adding NH_4OH, if M^{+2} and M^{+3} both the ions are in solution ?

(A) M^{+2}
(B) M^{+3}
(C) Both M^{+2} and M^{+3} together
(D) Precipitation will not take place.

Q68. For preparing a buffer solution of pH 6 by mixing sodium acetate and acetic acid, the ratio of concentration of salt and acid ($K_a = 10^{-5}$) should be

(A) 1 : 10 (B) 10 : 1
(C) 100 : 1 (D) 1 : 100.

Q69. The pH of an aqueous solution of $Ba(OH)_2$ is 10. If the K_{sp} of $Ba(OH)_2$ is 1×10^{-9}, then the concentration of Ba^{+2} ions in the solution is :

(A) 4×10^{-4} M (B) 5×10^{-6} M
(C) 2×10^{-2} M (D) 1×10^{-1} M

Q70. In solubility of salts M_2X, QY_2 and PZ_2 equal, then the relation between their K_{sp} will be :-

(A) $K_{sp}(M_2X) > K_{sp}(QY_2) > K_{sp}(PZ_2)$
(B) $K_{sp}(M_2X) = K_{sp}(QY_2) < K_{sp}(PZ_2)$
(C) $K_{sp}(M_2X) > K_{sp}(QY_2) = K_{sp}(PZ_2)$
(D) $K_{sp}(M_2X) = K_{sp}(QY_2) = K_{sp}(PZ_2)$

Q71. If 's' and 'S' are respectively solubility and solubility product of a sparingly soluble binary electrolyte then :–

(A) $s = S$ (B) $s = S^2$
(C) $s = S^{1/2}$ (D) $s = \frac{1}{2}S$

Q72. To precipitate Zn in form of ZnS, Why NH_4OH is first added in the solution before H_2S gas is passed through it :-

(A) To convert Zn into Zn^{+2}
(B) To reduce Zinc
(C) To decrease the dissociation of H_2S
(D) To increase the dissociation of H_2S

Q73. The solubility of calcium arsenate in pure water at 25°C is 9.0×10^{-5} mol/ lit. The solubility product, K_{sp} at this temperature is

(A) 8.1×10^{-9} (B) 6.4×10^{-20}
(C) 7.2×10^{-19} (D) 6.4×10^{-19}

Important JEE Advanced 2018 Problem

Q74. At 30°C, In which of the one litre solution, the solubility of Ag_2CO_3 (solubility product $= 8 \times 10^{-12}$) will be maximum :-

(A) 0.05 M Na_2CO_3 (B) Pure water
(C) 0.05 M $AgNO_3$ (D) 0.05 M NH_3

Q75. The solubility product constant Ksp of $Mg(OH)_2$ is 9.0×10^{-12}. If a solution is 0.010 M with respect to Mg^{2+} ion, what is the maximum hydroxide ion concentration which could be present without causing the precipitation of $Mg(OH)_2$:-

(A) 1.5×10^{-7} M (B) 3.0×10^{-7} M

(C) 1.5×10^{-5} M (D) 3.0×10^{-5} M

ACC

Q76. A sample of AgCl was treated with 5.0 M Na_2CO_3 solution to give Ag_2CO_3. The remaining solution contained 0.0026 gm of Cl^- per litre. Calculate solubility product of AgCl $(K_{sp(Ag_2CO_3} = 8.2 \times 10^{-12})$

(A) 0.71×10^{-10} (B) 2.12×10^{-10}

(C) 1.71×10^{-10} (D) 4.11×10^{-1}

Q77. What is the $[NH_4^+]$ in a solution of 0.02 M NH_3 and 0.01 M KOH (K_b of $NH_3 = 1.8 \times 10^{-5}$) ?

(A) 3.6×10^{-5} M

(B) 1.8×10^{-5} M

(C) 0.09×10^{-5} M

(D) 7.2×10^{-5} M

Q78. The dissociation constants of two acids HA_1 and HA_2 are 3.0×10^{-4} and 1.8×10^{-5} respectively.

The relative strengths of the acids will be approximately:-

(A) 1 : 4 (B) 4 : 1

(C) 1 : 16 (D) 16 : 1

Q79. A white substance having alkaline nature in solution is :-

(A) $NaNO_3$ (B) NH_4Cl

(C) Na_2CO_3 (D) Fe_2O_3

Q80. $HCOO^- + H_2O \square HCOOH + OH^-$ is related :-

(A) $h = \sqrt{K_h}$ (B) $h = \sqrt{\frac{K_h}{C}}$

(C) $h = \sqrt{\frac{K_h}{V}}$ (D) $K_h = \sqrt{hc}$

Q81. When equal volumes of the following solution are mixed, the precipitation of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) will occur with :

(A) 10^{-4} M (Ag^+) and 10^{-4} M (Cl^-)

(B) 10^{-5} M (Ag^+) and 10^{-5} M (Cl^-)

(C) 10^{-5} M (Ag^+) and 10^{-6} M (Cl^-)

(D) 10^{-10} M (Ag^+) and 10^{-10} M (Cl^-)

Q82. Buffer capacity is defined as the amount of acid or base that must be added to the buffer to cause :

(A) One unit change in pH

(B) Two units change in pH

(C) No Change in pH.

(D) pH equal to $p^{ka} \pm \left(\frac{1}{2}\right)$

Q83. If it is known that H_2S is a weak acid and it is ionised into $2H^+$ and S^{-2}. Then in this solution HCl is added so, pH becomes less, then what will happen :-

(A) Decrease in S^{-2} ion concentration

(B) Concentration of S^{-2} is not affected

(C) Increase in S^{-2} ion concentration

(D) It is not possible, to add HCl in solution

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The conjugate acid of NH_2^- is-

(A) NH_3 (B) NH_2OH

(C) NH_4^+ (D) N_2H_4

Q2. Which salt is not an example of acidic salt:-

(A) HCOONa

(B) NaH_2PO_2

(C) NaHS

(D) (A) and (B) both

Q3. pH values of two acids A and B are 4 and 5. The strengths of these two acids are related as :-

(A) The strengths of the two acids cannot be compared.

(B) Acid B is 10 times stronger than acid A.

(C) Strength of acid A : Strength of acid B = 4 : 5

(D) Acid A is 10 times stronger than acid B.

Q4. The ionization constant of acetic acid is 1.74×10^{-5}. Consider 0.05 M solution of acetic acid.

Choose the correct options –

(A) The concentration of acetate ion in the solution is 9.3×10^{-4} mol/L

(B) Concentration of hydrogen ion is 9.3×10^{-4} mol/L

(C) pH = 3.03

(D) pH = 7.03

Q5. The solubility of silver acetate in pure water at 25°C is 8.35 g L^{-1} and 61.8 g L^{-1} in an acid solution of pH = 3. Choose the correct options –

(A) K_{sp} of silver acetate is 2.5×10^{-3}

(B) Dissociation constant of acetic acid is 1.8×10^{-5}

(C) K_{sp} of silver acetate is 2.5×10^{-2}

(D) Dissociation constant of acetic acid is 1.8×10^{-3}

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 6 to 8

Acid rain is an environmental concern all over the world. In assessing the acidity of rainfall, it is important to have an idea of the acidity of natural rain water. Assuming that natural rain water (that is, rain water uncontaminated with nitric acid or sulphuric acid) is in equilibrium with 3.6×10^{-4} atm CO_2 (the Henry's law constant is 1.25×10^6 Torr) $K_{a_1}(H_2CO_3) = 4.3 \times 10^{-7}$

Q6. What is the pH of natural rain water ?

(A) 5.64 (B) 5.70

(C) 7.00 (D) 7.40

Q7. pH of natural rain water would have been in pre-industrial times when the partial pressure of CO_2 was about 2.8×10^{-4} atm-

(A) 7.0 (B) 5.64

(C) 7.4 (D) 5.70

Q8. If SO_2 content in the atmosphere is 0.12 ppm by volume, pH of rain water (assume 100% ionization of acid rain as monobasic acid) -

(A) 5.7 (B) 6.6

(C) 5.4 (D) 2.0

Paragraph for Q. 9 to 11

Solution of an acid and it's anion (that is, it's conjugate base) or of a base and it's common cation are buffered. When we add a small amount of acid or base to any one of them, the pH of solution changes very little. pH of buffer solution can be computed as

For acidic buffer : $pH = pK_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$

For basic buffer : $pOH = pK_b + \log \frac{[\text{conjugate base}]}{[\text{base}]}$

It is generally accepted that a solution has useful buffer capacity (pH change resisting power) provided that the value of [salt or conjugate base] / [acid] for acidic buffer lies within the range of 1 : 10 to 10 : 1. Buffer capacity is maximum when [conjugate base] = [acid]

Q9. Select incorrect statement -

(A) when we add small amount of NaOH in acidic buffer solution, pOH is increased

(B) when we add small amount of NaOH in basic buffer solution, pH of solution is increased

(C) when we add small amount of water in acidic buffer solution, pH of solution is decreases

(D) when 100 ml of 0.2 M CH_3COOH react with 200 ml of 0.1 M NaOH buffer solution is formed.

Q10. Which of the following mixtures constitute a buffer-

(A) HCOOH + HCOONa

(B) Na_2CO_3 + $NaHCO_3$

(C) NaCl + HCl

(D) NH_4Cl + $(NH_4)_2SO_4$

Q11. Fixed volume of 0.1 M benzoic acid (pK_a = 4.2) solution is added into 0.2 M sodium benzoate solution & formed a 300 mL, resultant acidic buffer solution. If pH of this buffer solution is 4.5 then find added volume of benzoic acid -

(A) 1 litre (B) 200 ml

(C) 2 litre (D) 0.20 litre

Paragraph for Q. 12 to 14

The property of resisting change of pH when an acid or alkali is added to it and this property of solution is known as buffer action of buffer solution may be acidic or basic depending upon the acid and base nature respectively pH calculation of buffer is given by Henderson equation.

$pH = pK_a + \log \frac{[\text{conjugate base}]}{[\text{acid}]}$ for acidic buffer

and

$pOH = pK_b + \log \frac{[\text{conjugate acid}]}{[\text{base}]}$ for basic buffer

Q12. 2gm of acetate ion in form of sodium acetate is added to 500ml of a solution of 0.10 M acetic acid pK_a (CH_3COOH) = 4.74 pH of solution is -

(A) 4.57 (B) 4.74

(C) 5.01 (D) 5.70

Q13. pH of solution containing equimolar amount of CH_3COOH (pK_a = 4.74) and CH_3COO at 50°C is -

(A) Less than 4.74 (B) More than 4.74

(C) Equal to 4.74 (D) None of these

Q14. The $[HCO_3^- / H_2CO_3]$ ratio in the blood (pH = 7.4) is approximately

(pKa = 6.1 for H_2CO_3)

(A) 20 (B) 16

(C) 24 (D) 10

Paragraph for Q. 15 to 17

When sparingly soluble salt is dissolved in water an equilibrium is established saturation is said to be attained at that temperature. The rate of dissolution of solid is equal to the rate of precipitation.

$AgCl(s) \rightleftharpoons Ag^{+}(aq) + Cl^{-}(aq)$

Equilibrium constant $K_{eq} = \dfrac{[Ag^{+}][Cl^{-}]}{[AgCl]}$

As concentration of AgCl undissociated is almost constant

$K_{eq}[AgCl] = [Ag^{+}][Cl^{-}]$, $K_{sp} = [Ag^{+}][Cl^{-}]$

Q15. A solution is a mixture at 0.05M NaCl and 0.05 M NaI. The concentration of iodide ion in the solution when AgCl is just starts precipitating is equal to -
(K_{sp} AgCl = 1×10^{-10} M^2,
K_{sp} AgI = 4×10^{-16} M^2)
(A) 4×10^{-6} M
(B) 2×10^{-8} M
(C) 2×10^{-7} M
(D) 8×10^{-15} M

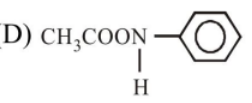
Q16. $M(OH)_x$ has $K_{sp} = 4 \times 10^{-12}$ Mand solubility 10^{-4} M. Hence x is –
(A) 1 (B) 2
(C) 3 (D) 4

Q17. Slaked lime $Ca(OH)_2$ is used extensively in sewage treatment. What is the maximum pH that can be established in $Ca(OH)_2$ (aq).
$Ca(OH)_2$ (s) $\rightleftharpoons Ca^{2+}$ (aq) + $2OH^{-}$ (aq),
$K_{sp} = 5.5 \times 10^{-6}$
(A) 1.66 (B) 12.35
(C) 7 (D) 14

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q),(R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q18. Match the expression for pH or degree of hydrolysis in column II with the nature of the salt in column I.

Column I	**Column II**
(A) KCN	(P) $h = \sqrt{\dfrac{K_w}{K_a \times C}}$
(B) CH_3COONa	(Q) $h = \sqrt{\dfrac{K_w}{K_b \times C}}$
(C) NH_4NO_3	(R) $pH = 7 - \dfrac{1}{2}pK_b - \dfrac{1}{2}\log C$
(D) 	(S) $pH = 7 + \dfrac{1}{2}pK_a + \dfrac{1}{2}\log C$
	(T) $pH = \dfrac{1}{2}(pK_w + pK_b - pK_a)$

Q19.

Column-I	Column-II
(A) NaCl	(P) Cationic Hydrolysis
(B) Na_2CO_3	(Q) Anionic Hydrolysis
(C) NH_4Cl	(R) pH = 7.0 at 25°
(D) CH_3COONH_4	(S) pH < 7.0 at 25°
	(T) pH > 7.0 at 25°
	(U) pH depends on pKa & pKb of weak acid & weak base

Q20.

Column-I	Column-II
(A) Buffer capacity is Maximum for an acid buffer is	(P) $pH = pKa + \log\frac{1}{3}$
(B) At the $\frac{1}{4}$ th of the equivalence point for the titration of weak acid HA	(Q) $\frac{1}{2}(pKa + \log 3)$
(C) At the $\frac{3}{4}$ th of the equivalence point for the titration of weak acid HA	(R) $pH = pKa - \log\frac{1}{3}$
(D) The pH of $\frac{1}{3}$ (M) solution of weak acid HA	(S) pH = pKa

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q21. When 2 moles of HCl is added to 1 L. of an acidic buffer solution, its pH changes from 3.9 to 3.4. Find its buffer capacity.

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q22. **Statement 1 :** It is difficult to compare the strengths of the strong acids such as HCl, H_2SO_4, HNO_3, HBr, HI or $HClO_4$ in dilute aqueous solutions.

Statement 2 : In dilute aqueous solution all strong acids donate H^+ ions to water molecules and are essentially 100% ionised to produce a solution containing H_3O^+ ions plus the anions of strong acid.

Q23. **Statement - 1 :** The addition of silver ions to a mixture of aqueous sodium chloride and sodium bromide solution, will first precipitate AgBr rather than AgCl.

Statement - 2 : The value of K_{sp} of AgCl < K_{sp} of AgBr.

SUBJECTIVE TYPE

Q24. Find out the value of α of 10^{-2} M HCN solution if $[H^+] = 10^{-3}$.

Q25. For 10 M CH_3COOH solution if $K_a = 10^{-5}$ then find out :

(i) α (ii) $[H^+]$ (iii) pH

Q26. How does Lewis theory explain the acidic character of CO_2 ?

Q27. What should be the number of H^+ ions in 1 mL of distilled water, if the number of H^+ ions in 1 L is 6.023×10^{16} ?

Q28. What will be the pH of a solution obtained by mixing 800 mL of 0.05 N sodium hydroxide and 200 mL of 0.10 N HCl, assming complete dissociation of acid & base?

Q29. Find out the solubility of AgCl in the presence of M NaCl solution ?

Q30. Find out the solubility of $CaCl_2$ solution in the presence of M NaCl solution ?

Q31. (a) Will $Mg(OH)_2$ precipitate from a solution that is 0.01 M in $MgCl_2$ and also 0.10 M in NH_3.

(b) What NH_4^+ must maintained to prevent the precipitation of $Mg(OH)_2$ from a solution that is 0.010 $MgCl_2$ in 0.1 M NH_3 K_{sp} Mg $(OH)_2 = 2.5 \times 10^{-11}$, K_b $NH_3 = 1.8 \times 10^{-5}$.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Find out K_a for 10^{-2} M HCN acid, having pOH is 10 :-

(A) $K_a = 10^{-4}$ (B) $K_a = 10^{-2}$

(C) $K_a = 10^{-5}$ (D) $K_a = 10^{-6}$

Q2. If 0.1 M of a weak acid is taken and its of degree of ionization is 1.34%, then calculate its ionization constant :-

(A) 0.8×10^{-5} (B) 1.79×10^{-5}

(C) 0.182×10^{-5} (D) None of these

Q3. Correct statement for HCN weak acid at 25^0C temperature.:-

A (a) $\alpha = \dfrac{K_a}{[H^+]}$ (b) $\alpha = \dfrac{K_a \times [OH^-]}{K_w}$

(C) (A) & (B) both (D) $K_b = Ca^2$

Q4. If α is the degree of dissociation of weak dibasic organic acid and y is the hydrogen ion concentration, what is the initial concentration of acid :-

(A) $\dfrac{\alpha(y)^{-1}}{2}$ M (B) $y(\alpha)^{-1}$M

(C) $\dfrac{y(\alpha)^{-1}}{2}$ M (D) None of them

Q5. If acetic acid 5% ionized in 0.1 M solution, then its pH value is -

(A) 2.31 (B) 2.71

(C) 2.91 (D) 3.01

Q6. Three reactions involving $H_2PO_4^-$ are given below :-

(i) $H_3PO_4 + H_2O \rightarrow H_3O^+ + H_2PO_4^-$

(ii) $H_2PO_4^- + H_2O \rightarrow HPO_4^{2-} + H_3O^+$

(iii) $H_2PO_4^- + OH^- \rightarrow H_3PO_4 + O^{2-}$

In which of the above does $H_2PO_4^-$ act as an acid ?

(A) (i) only (B) (ii) only

(C) (i) and (ii) (D) (iii) only

Q7. A solution of $Na_2S_2O_3$ is standardized iodometrically against 0.167 g of $KBrO_3$ where BrO_3^- changes to Br^-. This process requires 45 mL of the $Na_2S_2O_3$ solution. What is the strength of the $Na_2S_2O_3$?

(A) $\frac{2}{15}$N (B) $\frac{2}{30}$N

(C) $\frac{1}{30}$N (D) $\frac{1}{60}$N

Q8. Equal volume of two solutions of pH = 3 and 4 are mixed the pH of resulting solution will be :-

(A) 7 (B) 3.5

(C) 2.96 (D) 3.26

Q9. Equal volume of two solutions of pH = 3 and 4 are mixed the pH of resulting solution will be :-

(A) 7 (B) 3.5

(C) 2.96 (D) 3.26

Q10. The pH of a 10^{-9} M NaOH solution is nearest to :-

(A) 5 (B) 9

(C) 7 (D) –10

Q11. What is the $[OH^-]$ in the final solution prepared by mixing 20 ml,0.5 MHCl with 30 ml of 0.1M $Ba(OH)_2$?

(A) 0.1 M

(B) 0.4 M

(C) 0.005 M

(D) 0.08 M

Q12. Calculate the pH of a solution obtained by mixing 2 ml of HCl of pH 2 and 3 ml of solution of KOH of pH = 12 :-

(A) 10.30 (B) 3.70

(C) 11.30 (D) None

Q13. Which of the following has pH value of 14?

(A) 1 M HCl (B) 1 M NaOH

(C) 0.1 M NaOH (D) 0.1 M H_2SO_4

Q14. At 25°C, the dissociation constant of a base BOH is 1.0×10^{-12}. The concentration of Hydroxyl ions in 0.01 M aqueous solution of the base would be:-

(A) 2.0×10^{-6} mol L^{-1}

(B) 1.0×10^{-5} mol L^{-1}

(C) 1.0×10^{-6} mol L^{-1}

(D) 1.0×10^{-7} mol L^{-1}

Q15. Equilibrium constants of T_2O (T or $_1H^3$ is an isotope of $_1H^1$) and H_2O are different at 298 K. Let at 298K pure T_2O has pT (like pH) is 7.62. The pT of a solution prepared by adding 10 mL of 0.2M TCl to 15 mL of 0.25 M NaOT is :-

(A) 12 – log 7 (B) 12 + log 7

(C) 13.24 – log 7 (D) 13.24 + log 7

Q16. Choose the wrong statement :-

(A) For a neutral solution :

$[H^+] = [OH^-] = \sqrt{K_w}$

(B) For an acidic solution :

$[H^+] > \sqrt{K_w}$ and $[OH^-] < \sqrt{K_w}$

(C) For a basic solution :

$[H^+] < \sqrt{K_w}$ and $[OH^-] > \sqrt{K_w}$

(D) For a neutral solution at all temperatures:

$[H^+] = [OH^-] = 10^{-7}$ M

Q17. The pH of a soft drink is 3.82. The hydrogen ion concentration will be :

[given antilog 0.18 = 1.5]

(A) 1.96×10^{-2} mol lit^{-1}

(B) 1.96×10^{-5} mol lit^{-1}

(C) 1.5×10^{-4} mol lit^{-1}

(D) 1.5×10^{-2} mol lit^{-1}

Q18. Find out pH of solution having 2×10^{-3} moles of OH^- ion's in 2 litre solution :-

(A) pH = 3

(B) pH = 3 + log2

(C) pH = 3 – log2

(D) pH = 11

Q19. The pH of solution is increased from 3 to 6. Its H^+ ion concentration will be :-
(A) Reduced to half
(B) Doubled
(C) Reduced by 1000 times
(D) Increased by 1000 times

Q20. 9.8 g of H_2SO_4 is present in 500 ml of solution. Calculate the pH of the solution :-
(A) 0.99 (B) 2
(C) 0.2010 (D) 0.4

Q21. pH of mono basic weak acid can be given by :-
(A) $-\log[H^+]$ (B) $-\log[\alpha C]$
(C) $\frac{1}{2}pK_a - \frac{1}{2}\log C$ (D) All of these

Q22. Which of the following is right for diprotic acid :
(A) $Ka_2 > Ka_1$ (B) $Ka_1 > Ka_2$
(C) $Ka_2 > \frac{1}{Ka_1}$ (D) $Ka_2 = Ka_1$

Q23. A 50 ml solution of pH = 1 is mixed with a 50 ml solution of pH = 2. The pH of the mixture will be nearly :-
(A) 0.76 (B) 1.26
(C) 1.76 (D) 2.26

Q24. If pK_b for CN^- at 25°C is 4.7. The pH of 0.5M aqueous NaCN solution is :-
(A) 12 (B) 10
(C) 11.5 (D) 11

Q25. On dilution, pH of CH_3COONa :-
(A) Decreases
(B) Increases
(C) Remains unchanged
(D) First increases then decreases

Q26. The number of OH^- ions in 1 mL of solution having pH = 4 is :-
(A) 10^{-4} (B) 10^{-10}
(C) 6.02×10^{10} (D) 6.02×10^{13}

Q27. For which salt relation between K_h and h $h = \frac{\sqrt{K_h}}{1+\sqrt{K_h}}$ true :-
(A) KCN (B) NH_4NO_3
(C) CH_3COONH_4 (D) $KClO_4$

Q28. Ammonium cyanide is a salt of NH_4OH of $K_b = 1.8 \times 10^{-5}$ and HCN of $K_a = 4.0 \times 10^{-10}$. The hydrolysis constant of NH_4CN at 25°C is -
(A) 1.4 (B) 2.4 (C) 3.4 (D) 4.6

Q29. Hydrolysis constant of CH_3COONH_4 is proportional to the following :-
(A) Kw^{-1} (B) $(K_a \times K_b)^{-1}$
(C) $K_a - K_b$ (D) $\sqrt{K_w}$

Q30. What should be the concentration of base in 0.1 M NH_4Cl solution, it its $K_b = 10^{-5}$:-
(A) 10^{-7} (B) 10^{-8} (C) 10^{-6} (D) 10^{-5}

Q31. What will be the pH and % h (degree of hydrolysis) respectively for the salt BA of 0.1 M concentration ?
Given : K_a for HA = 10^{-6} and K_b for BOH = 10^{-6} :
(A) 5, 1 % (B) 7, 10%
(C) 9, 0.01 (D) 7, 0.001 %

Q32. For anionic hydrolysis, pH is given by:-
(A) $pH = \frac{1}{2}pK_W - \frac{1}{2}pK_b - \frac{1}{2}\log c$
(B) $pH = \frac{1}{2}pK_W + \frac{1}{2}pK_a - \frac{1}{2}pK_b$
(C) $pH = \frac{1}{2}pK_W + \frac{1}{2}pK_a + \frac{1}{2}\log c$
(D) None of above

Q33. Four solutions of NH_4Cl are taken with concentrations 1M, .01M, .01M and .001M. Their degree of hydrolysis are h_1, h_2, h_3 and h_4. What is order of degree of hydrolysis :-
(A) $h_1 > h_2 > h_3 > h_4$
(B) $h_4 > h_3 > h_2 > h_1$
(C) $h_1 = h_2 = h_3 = h_4$
(D) $h_4 > h_3 = h_2 > h_1$

Q34. Handerson equation $pH - pK_a = 1$ will be applicable to an acidic buffer when :-
(A) [Acid] = [Conjugate base]
(B) [Acid] × 10 = [Conjugate base]
(C) [Acid] = [Conjugate base] × 10
(D) [Acid] = [Conjugate base] × 0.1

Q35. On addition of NaOH to CH_3COOH solution, 60% of the acid is neutralised. If pK_a of CH_3COOH is 4.7 then the pH of the resulting solution is :-
(A) More than 4.7 but less than 5.0
(B) Less than 4.7 but more than 4.0
(C) More than 5.0
(D) Remains unchanged

Q36. pH of a solution obtained by mixing equal volume of 0.2 M NaOH & 0.2 M CH_3COOH ($K_a = 10^{-5}$) is :
(A) 7 (B) 5 (C) 9 (D) 9.5

Q37. If the equilibrium constant of the reaction of weak acid HA with NaOH is 10^9, then pH of 0.1 M NaA solution is :-
(A) 5 (B) 9 (C) 7 (D) 8

Q38. Which of the following solutions does not act as buffer :–
(A) $H_3PO_4 + NaH_2PO_4$
(B) $NaHCO_3 + H_2CO_3$
(C) $NH_4Cl + HCl$
(D) $CH_3COOH + CH_3COONa$

Q39. When 1.0 mL of dil. HCl acid is added to 100 mL of a buffer solution of pH 4.0. The pH of the solution.
(A) Becomes 7 (B) Does not change
(C) Becomes 2 (D) Becomes 10

Q40. A basic - buffer will obey the equation $pOH - pK_b = 1$ only under condition :-
(A) [Conjugate acid] : [base] = 1 : 10
(B) [Conjugate acid] = [base]
(C) [Conjugate acid] : [base] = 10 : 1
(D) N.O.T

Q41. An acid-base indicator which is a weak acid has a pK_a value = 5.5. At what concentration ratio of sodium acetate to acetic acid would the indicator show a colour half-way between those of its acid and conjugate base forms ? pK_a of acetic acid = 4.75
(A) 4.93 : 1 (B) 6.3 : 1
(C) 5.62 : 1 (D) 2.37 : 1

Q42. The rapid change of pH near the stoichiometric point of an acid–base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentrations of the conjugate acid (HIn) and base (In^-) forms of the indicator by the expression:-
(A) $\log \frac{[HIn]}{[In^-]} = pK_{In} - pH$
(B) $\log \frac{[HIn]}{[In^-]} = pH - pK_{In}$
(C) $\log \frac{[In^-]}{[HIn]} = pH + pK_{In}$
(D) $\log \frac{[In^-]}{[HIn]} = pK_{In} - pH$

Q43. Why only As^{3+} gets precipitated as As_2S_3 and not Zn^{2+} as ZnS when H_2S is passed through an acidic solution containing As^{3+} and Zn^{2+} :-
(A) Enough As^{3+} are present in acidic medium
(B) Zinc salt does not ionise in acidic medium
(C) Solubility product of As_2S_3 is less than that of ZnS
(D) Solubility product changes in presence of an acid

Q44. M_2SO_4 (M^+ is a monovalent metal ion) has a K_{sp} of 1.2×10^{-5} at 298 K. The maximum concentration of M^+ ion that could be attained in a saturated solution of this solid at 298 K is :-
(A) 3.46×10^{-3} M (B) 2.89×10^{-2} M
(C) 2.8×10^{-3} M (D) 7.0×10^{-3} M

Q45. If the solubility of $PbBr_2$ is 'S' g molecules per litre, considering 100% ionisation its solubility product is :-

(A) $2S^3$ (B) $4S^2$

(C) $4S^3$ (D) $2S^4$

Q46. The solubility of $BaSO_4$ in water, is 2.33×10^{-3} gL^{-1}. Its solubility product will be (molecular weight of $BaSO_4$ = 233) :-

(A) 1×10^{-5} (B) 1×10^{-10}

(C) 1×10^{-15} (D) 1×10^{-20}

Q47. If the solubility of AgCl (formula mass=143) in water at 25°C is 1.43×10^{-4} g/100 mL of solution then the value of K_{sp} will be :-

(A) 1×10^{-5} (B) 2×10^{-5}

(C) 1×10^{-10} (D) 2×10^{-10}

Q48. The solubility product of $BaSO_4$ at 25°C is 1.0×10^{-9}. What would be the concentration of H_2SO_4 necessary to precipitate $BaSO_4$ from a solution of 0.01 M Ba^{+2} ions :-

(A) 10^{-9} (B) 10^{-8}

(C) 10^{-7} (D) 10^{-6}

Q49. Concentration of Ag^+ ions in saturated solution of Ag_2CrO_4 at 20°C is 1.5×10^{-4} mol L^{-1}. At 20°C, the solubility product of Ag_2CrO_4 is :-

(A) 3.3750×10^{-12}

(B) 1.6875×10^{-10}

(C) 1.68×10^{-12}

(D) 1.6875×10^{-11}

Q50. What will happen if the pH of the solution of 0.001 M $Mg(NO_3)_2$ solution is adjusted to pH = 9 ($K_{sp}Mg(OH)_2 = 8.9 \times 10^{-12}$)

(A) ppt will take place

(B) ppt will not take place

(C) Solution will be saturated

(D) None of these

Q51. The dissociation constant for acetic acid and HCN at 25°C are 1.5×10^{-5} and 4.5×10^{-10} respectively. The equilibrium constant for the equilibrium $CN^- + CH_3COOH \rightleftharpoons HCN + CH_3COO^-$ would be :-

(A) 3.0×10^{-5} (B) 3.3×10^{-4}

(C) 3.3×10^{4} (D) 3.0×10^{5}

Q52. In the system of a saturated solution of CaF_2 dissolved in water at a given temperature

$CaF_2(s) \rightleftharpoons \underset{(aq.)}{Ca^{2+}} + 2\underset{(aq.)}{F^-}$ adding a calcium salt increases the concentration of Ca^{2+} ions to n times.

Concentration of F^- will become :-

(A) n times the previous value

(B) 1/n times of the previous value

(C) $\frac{1}{\sqrt{n}}$ times of the previous value

(D) $\sqrt{n}$ times of previous value

Q53. Which pair will show common ion effect :-

(A) $BaCl_2 + Ba(NO_3)_2$

(B) NaCl + HCl

(C) $NH_4OH + NH_4Cl$

(D) AgCN + KCN

Q54. Which of the following is a true statement:

(A) The ionisation constant and ionic product of water are same.

(B) Water is a strong electrolyte.

(C) The value of ionic product of water is less than that of its ionisation constant.

(D) At 298K, the number of H^+ ions in a litre of water is 6.023×10^{16}.

Q55. The pH values of 0.1 solution of HCOONa (I), HCOOH (II), CH_3COONH_4(III), NaOH (IV), HCl (V) will be in the order :-

(A) IV > III > I > II > V

(B) IV > I > III > II > V

(C) II > III > I > IV > V

(D) V > II > III > I > IV

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. 2.7 g weak acid HA (mol. wt. = 27 g/mol.) is dissolved in water and formed 10 m^3 solution. If $K_{a(HA)} = 10^{-9}$ then pOH of solution will be [**Given : log 4 = 0.6**]

(A) 6.7

(B) Greater than 6.7 & less than 7.0

(C) Greater than 7.0 & less than 7.3

(D) Greater than 7.3

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions.Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 2 to 4

A solution of paranitrophenol indicator of concentration 6×10^{-4} (M) was prepared and indicator solution was used in the spectrophotometric measurement. The experimental results are tabulated as below

Condition	Form of Indicator	Absorbance
Strongly Acidic	HIn	0.142
Strongly Alkaline	In^-	0.943
pH = 8	$HIn + In^-$	0.527

Given : log 2 = 0.3 ; log 3 = 0.48 ;
log 5 = 0.7 ; log 7 = 0.85

Q2. What is the molar concentration of In^- at pH = 8 ?

(A) 3.11×10^{-4} (M) (B) 2.88×10^{-4} (M)

(C) 6×10^{-4} (M) (D) None of these

Q3. What is the K_{in} of the indicator ?

(A) 3.117×10^{-4} (B) 6.12×10^{-9}

(C) 9.25×10^{-10} (D) 9.25×10^{-9}

Q4. Which of the following is/are true regarding the choice of indicator ?

(A) At the neutral point of the indicator $[In^-] = [HIn]$

(B) At the equivalence point of acid base titration entire amount of acid is neutralized by the base

(C) The equivalence point may or may not be equal to the neutral point of indicator

(D) Best indicator to identify the end point of the acid base titration is such indicator for which pH corresponding to equivalence point lie within the $pK_{in} \pm 1$.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q5.

Column - I	Column - II
(A) Bronsted-Lowry acid	(P) CH_3^+
(B) Bronsted Lowry base	(Q) CH_3^-
(C) Lewis acid	(R) $H_2PO_4^-$
(D) Lewis base	(S) NH_4^+

Q6. K_a and K_b are the dissociation constant of weak acid and weak base and K_w is the ionic product of water

Column I	**Column II**
(A) 0.1(M) KCN solution	(P) pH $\simeq$ 7
(B) 0.1(M) $C_6H_5NH_3Cl$ solution	(Q) $pH = 6.5 + \frac{1}{2}pK_a$
(C) 0.1(M) KCl solution	(R) $pH = 7.5 - \frac{1}{2}pK_b$
(D) 0.1(M) CH_3COONH_4 solution	(S) $pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q7. **Statement 1 :** $HClO_4$ acidic strength is more than that of $HClO_3$.
Statement 2 : In ClO_4^- negative charge density is less than that of ClO_3^-

Q8. **Statement 1 :** Weak acid + Salt of this acid with a strong base will behave like a buffer.
Statement 2 : Weak base + Salt of this base with a strong acid behave like a buffer.

SUBJECTIVE TYPE

Q9. Calculate the pH of 2.0×10^{-4} M H_3X Solution assuming first dissociation to be 100%, second to be 50% and third to be negligible :

Q10. A weak acid (50.0 mL) was titrated with 0.1 M NaOH. The pH values when 10.0 mL and 25.0 mL of base have been added are found to be 4.16 and 4.76, respectively. Calculate K_a of the acid and pH at the equivalence point.

Q11. Write the equation for K_{sp} ?

(i) $Al_2(SO_4)_3(s) \rightleftharpoons 2Al^{+3}(aq) + 3SO_4^{-2}(aq)$

(ii) $Na_2KPO_4(s) \rightleftharpoons 2Na^+(aq) + K^+(aq) + PO_4^{-3}(aq)$

(iii) $NaKRbPO_4(s) \rightleftharpoons Na^+(aq) + K^+(aq) + Rb^+(aq) + PO_4^{-3}(aq)$

Q12. The solubility product of AgBr is 5.2×10^{-13}. Calculate its solubility in mol dm^{-3} and g dm^{-3}. (Molar mass of AgBr. = 187.8 g mol^{-1})

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Column I	Column II	
(A) Bronsted-Lowry acid	(P) CH_3^+	(i) Cationic hydrolysis
(B) Bronsted Lowry base	(Q) CH_3^-	(ii) Anionic hydrolysis
(C) Lewis acid	(R) $H_2PO_4^-$	(iii) Amphiprotic ion
(D) Lewis base	(S) NH_4^+	(iv) Dibasic

Q13. Which of the following is correct match :
(A) A, R, iii (B) A, R, i (C) D, Q, i (D) None

Q14. Which of the following is correct match :
(A) B, R, iv (B) B, R, i (C) C, R, iv (D) D, R, iii

Q15. Which of the following is correct match :
(A) C, S, i (B) C, S, iv (C) C, P, i (D) D, Q, i

Answer Key

EX-1									
1	2	3	4	5	6	7	8	9	10
C	D	C	A	D	C	D	A	A	C
11	12	13	14	15	16	17	18	19	20
D	B	B	C	A	D	B	A	A	C
21	22	23	24	25	26	27	28	29	30
B	B	C	D	A	B	B	D	C	B
31	32	33	34	35	36	37	38	39	40
B	D	D	A	B	C	C	B	D	C
41	42	43	44	45	46	47	48	49	50
B	C	B	A	D	B	D	B	C	C
51	52	53	54	55	56	57	58	59	60
C	C	C	C	B	A	C	A	D	C
61	62	63	64	65	66	67	68	69	70
D	C	C	B	C	A	B	B	D	D
71	72	73	74	75	76	77	78	79	80
C	D	D	D	D	C	A	B	C	B
81	82	83							
A	A	A							

Ex-1A

1. A **2.** D **3.** D **4.** A,B,C **5.** A,B
6. A **7.** D **8.** A **9.** A,C,D **10.** A,B
11. B,D **12.** A **13.** C **14.** A **15.** C
16. B **17.** B
18. A→ P,S ; B → P,S ; C → Q,R ; D → T
19. A → R ; B → Q,T ; C → P,S ; D → U
20. A → S ; B → P ; C → R; D → Q
21. 4 **22.** A **23.** C **24.** 10
25. (i) 10^{-3} (ii) 10^{-2} (iii) 2 **27.** 6.023×10^{13} **28.** 12.3
29. $\frac{Ksp}{C}$ **30.** $\frac{Ksp}{C^2}$ **31.** (a) yes (b) 0.036 M

Ex-2

1	2	3	4	5	6	7	8	9	10
D	B	C	C	A	B	A	D	D	C
11	12	13	14	15	16	17	18	19	20
D	C	B	D	D	D	C	D	C	D
21	22	23	24	25	26	27	28	29	30
D	B	B	C	A	C	C	A	B	D
31	32	33	34	35	36	37	38	39	40
B	C	D	B	A	C	B	C	B	C
41	42	43	44	45	46	47	48	49	50
C	A	C	B	C	B	C	D	C	B
51	52	53	54	55					
C	C	C	D	B					

Ex-2A

1. C **2.** B **3.** D **4.** D
5. A → R,S ; B → Q,R; C → P ; D → Q,R
6. A → Q; B → R; C → P; D → P,S]
7. A **8.** B **9.** 3.52 **10.** 8.73
11. (i) 108 S^5, (ii) $4S^4$, (iii) S^4 **12.** 1.35×10^{-4} g dm^{-3} & 7.2×10^{-7} mol dm^{-3}
13. A **14.** A **15.** A